Servo Arm Control System

User Manual

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1. Product Overview

This software controls a 5-servo robotic arm using joysticks or serial commands.

Compatible with Arduino UNO and similar boards.

2. Safety Warning

Power Supply:

- After uploading code, switch from USB to external power supply

- Use 5V 1A or higher power adapter

- This ensures stable operation for board and servos

Overload Protection:

- Do not apply excessive force when grasping objects

- If servo stalls, current increases sharply

- Overcurrent will trigger board protection and restart

3. Pin Definition

Joystick Pins:

Joystick 1 X-axis: A0 (Analog Pin 0)

4. Control Logic

Control Mapping Table:

Control	Pin	Servo	Range
Joystick 1 X	A0	Servo 0	0-180 deg
Joystick 1 Y	A1	Servo 1	0-180 deg
Joystick 1 Btn	D2	Servo 4	0-90 deg
Joystick 2 X	A3	Servo 2	0-180 deg
Joystick 2 Y	A2	Servo 3	0-180 deg
Joystick 2 Btn	D4	Servo 4	0-90 deg

Button Control:

- Joystick 1 Button: Servo 4 +2 deg per press (max 90 deg)

- Joystick 2 Button: Servo 4 -2 deg per press (min 0 deg)

5. Arduino Setup

1. Open servo_control.ino in Arduino IDE

2. Select your board (Arduino UNO/Nano)

7. Serial Commands (Continued)

Example:

> 0 90

S0:90

8. Python GUI Setup

1. Install Python 3.x
2. Install pyserial: pip install pyserial
3. Run: python servo_control.py
4. Select COM port and click Connect
5. Use sliders or enter values to control servos

9. Support

GitHub: https://github.com/490437477-sys/servo-arm-control